# 🎙️ SpeechToTextWidget Documentation

A reusable Flutter widget that converts **speech to text** with a glowing microphone animation. It uses the `speech_to_text` package for voice recognition and `avatar_glow` for visual feedback while listening.

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## Packages Used

```
dependencies:
  flutter:
    sdk: flutter
  speech_to_text: ^6.6.0
  avatar_glow: ^3.0.1
  flutter_screenutil: ^5.9.0
```

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## Features

- 🎤 Tap-to-start / tap-to-stop voice recording
- 🏅 Animated glowing mic while listening
- Real-time speech-to-text conversion
- 🎈 Callback with recognized text
- 🧩 Accepts initial text
- Responsive sizing with `ScreenUtil`

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## How It Works

1. Initializes speech recognition on widget load
2. Starts listening when microphone is tapped
3. Updates text live as user speaks
4. Sends recognized text back via callback
5. Stops listening on second tap or widget dispose

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## 🧩 Widget Parameters

| Parameter | Type | Description |
|---|---|---|
| `onResult` | `Function(String)` | Returns recognized speech text |
| `initialText` | `String?` | Pre-filled text (optional) |

## 🧨Example Usage

```
SpeechToTextWidget(
  initialText: "Say something...",
  onResult: (text) {
    print(text);
  },
)
```

## Full Source Code

```
import 'package:avatar_glow/avatar_glow.dart';
import 'package:flutter/material.dart';
import 'package:flutter_screenutil/flutter_screenutil.dart';
import 'package:isaacmwesigw352/assets_helper/app_colors.dart';
import 'package:speech_to_text/speech_to_text.dart';

class SpeechToTextWidget extends StatefulWidget {
  final Function(String text) onResult;
  final String? initialText;

  const SpeechToTextWidget({
    super.key,
    required this.onResult,
    this.initialText,
  });

  @override
  State<SpeechToTextWidget> createState() => _SpeechToTextWidgetState();
}

class _SpeechToTextWidgetState extends State<SpeechToTextWidget> {
  final SpeechToText _speech = SpeechToText();
  bool _isListening = false;
  String _text = '';

  @override
  void initState() {
    super.initState();
    _text = widget.initialText ?? '';
    _initSpeech();
  }
```

```
Future<void> _initSpeech() async {
  await _speech.initialize();
}

void _toggleListening() {
  if (_isListening) {
    _speech.stop();
    setState(() => _isListening = false);
  } else {
    _speech.listen(
      listenMode: ListenMode.dictation,
      onResult: (result) {
        setState(() {
          _text = result.recognizedWords;
        });
        widget.onResult(_text);
      },
    );
    setState(() => _isListening = true);
  }
}

@override
void dispose() {
  _speech.stop();
  super.dispose();
}

@override
Widget build(BuildContext context) {
  return Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [
      Text(
        _text.isEmpty ? '' : _text,
        style: const TextStyle(fontSize: 16),
      ),
      const SizedBox(height: 20),
      AvatarGlow(
        glowShape: BoxShape.circle,
        animate: _isListening,
        glowColor: AppColors.c0F86A4,
        curve: Curves.easeInCirc,
        duration: const Duration(milliseconds: 1000),
        repeat: true,
        child: SizedBox(
          width: 96.w,
```

```
            height: 96.w,
            child: Material(
              shape: const CircleBorder(),
              child: Ink(
                decoration: BoxDecoration(
                  gradient: LinearGradient(
                    colors: [
                      AppColors.c4F46E5.withAlpha(100),
                      AppColors.c3B82F6.withAlpha(100),
                    ],
                    begin: Alignment.topCenter,
                    end: Alignment.bottomCenter,
                  ),
                  shape: BoxShape.circle,
                ),
                child: InkWell(
                  customBorder: const CircleBorder(),
                  onTap: _toggleListening,
                  child: Center(
                    child: Icon(
                      _isListening ? Icons.mic : Icons.mic_none,
                      size: 28.sp,
                      color: Colors.white,
                    ),
                  ),
                ),
              ),
            ),
          ),
        ),
      ),
    ],
  );
 }
}
```

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## Notes

- Make sure microphone permission is added in **AndroidManifest.xml** and **Info.plist**
- Call `ScreenUtilInit()` in your app root
- Customize glow colors via `AppColors`

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## Use Cases

- Voice search

- Chat input
- Accessibility features
- Form voice input

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Happy Coding 🖌️